Exercises 10

Exercise 1 (QFT on 1 and 2 qubits)**.** *Recall that the matrix entries of the $M \times M$ quantum Fourier transform* QFT *are*

$$\mathrm{QFT}_{jk} = \omega^{jk}/\sqrt{M}$$

where $\omega = \exp(2\pi i/M)$ and $j, k \in \{0, \ldots, M-1\}$. Write down the matrix corresponding to the quantum Fourier transform on

1. *one qubit,*
2. *two qubits.*

Exercise 2 (Quantum circuit for QFT on 2 qubits)**.** *The goal of this exercise is to better understand the quantum Fourier transform on two qubits.*

1. *You saw the circuit for the 4-qubit quantum Fourier transform in the class. Draw the 2-qubit version of this circuit for the 2-qubit* QFT*.*
2. *Using this circuit, verify that*

$$\mathrm{QFT}\,|b_1 b_2\rangle = \frac{1}{2}\big(|0\rangle + (-1)^{b_2}|1\rangle\big) \otimes \big(|0\rangle + (-1)^{b_1} i^{b_2}|1\rangle\big).$$

3. *Verify that this agrees with the 2-qubit* QFT *matrix you obtained in the previous exercise.*

Exercise 3 (Roots of unity)**.** *Let $\omega = \exp(2\pi i/M)$ be the M-th root of unity.*

1. *Show that* $\displaystyle\sum_{k=0}^{M-1} \omega^k = 0.$
2. *Show that* $\displaystyle\sum_{k=0}^{M-1} \omega^{ak} = M$ *if* $a \equiv 0 \pmod{M}$.
3. *Show that* $\displaystyle\sum_{k=0}^{M-1} \omega^{ak} = 0$ *if* $a \not\equiv 0 \pmod{M}$.

Hint: *You can prove the first identity by contradiction: assume the sum is not 0 and multiply both sides by ω to get a contradiction. Alternatively, you can use the formula for the sum of the first n terms in a geometric progression: if $x \neq 1$ then $\sum_{k=0}^{n-1} x^k = (1-x^n)/(1-x)$.*

Exercise 4 (Unitarity of QFT). *Let* QFT *denote the $M \times M$ quantum Fourier transform. Recall that* $\mathrm{QFT}_{jk} = \omega^{jk}/\sqrt{M}$ *where* $\omega = \exp(2\pi i/M)$ *and* $j, k \in \{0, \ldots, M-1\}$. *Write down an expression for the matrix entry* $(\mathrm{QFT}^\dagger)_{kl}$ *and verify that* QFT *is unitary.*

Exercise 5 (Product representation of the QFT output). *Note that any number $b \in [0,1)$ that is of the form $a/2^n$, for some $a \in \{0, \ldots, 2^n - 1\}$, can be uniquely expressed as*

$$b = \sum_{i=1}^{n} b_i 2^{-i},$$

for some coefficients $b_1, \ldots, b_n \in \{0,1\}$. In fact, $0.b_1 b_2 \ldots b_n$ is simply the binary representation of b and b_k is the k-th digit of this representation.

For any $j \in \{1, \ldots, n\}$, let $0.b_j b_{j+1} \ldots b_n \in [0,1)$ be a real number whose binary representation agrees with the last binary digits of b.

1. *Show that $0.b_j b_{j+1} \ldots b_n$ is the binary representation of the fractional part of $2^{j-1}b$ (this is the part that comes after the point in the binary expansion).*

Let $\beta_j = \exp(2\pi i\, 0.b_j b_{j+1} \ldots b_n) \in \mathbb{C}$ be a complex number whose phase is $2\pi \cdot 0.b_j b_{j+1} \ldots b_n$.

2. *Show that $\beta_{n+1-k} = \exp(2\pi i b 2^{n-k})$.*

Recall that the n-qubit quantum Fourier transform is given by the following $2^n \times 2^n$ matrix:

$$\mathrm{QFT} = \frac{1}{\sqrt{2^n}} \sum_{a,b=0}^{2^n-1} \omega^{ab} |a\rangle\langle b|$$

where $\omega = \exp(2\pi i/2^n)$. The rest of this exercise is about proving the following identity:

$$\mathrm{QFT}\,|b_1 b_2 \ldots b_n\rangle = \frac{1}{\sqrt{2^n}} \big(|0\rangle + \beta_n |1\rangle\big) \otimes \big(|0\rangle + \beta_{n-1}|1\rangle\big) \otimes \cdots \otimes \big(|0\rangle + \beta_1 |1\rangle\big). \quad (1)$$

It can be proved by showing that the vectors on both sides agree entry-wise. For this, it suffices to left-multiply both sides with $\langle a_1 a_2 \ldots a_n|$ and compare the results. If you get the same result for all $a_1, a_2, \ldots, a_n \in \{0,1\}$ then all 2^n entries agree and hence eq. (1) holds.

3. *Show that the resulting entry on the left-hand side of eq. (1) is $\omega^{ab}/\sqrt{2^n}$.*
4. *Show that the resulting entry on the right-hand side of eq. (1) is $\frac{1}{\sqrt{2^n}} \prod_{k=1}^{n} \beta_{n+1-k}^{a_k}$.*
5. *Simplify this expression and show that both sides are equal.*